

B2 Symmetry and Relativity — Model Solutions, 2017

Question 1 — 4-vectors, pure force, annihilation

Problem. Define $\mathbf{U}$, $\mathbf{F}$; show orthogonal non-zero 4-vectors split into space-/time-like; show $\mathbf{P} \cdot \mathbf{P} = 0 \Rightarrow m = 0$. Define proper time, pure force; show $\mathbf{F} \cdot \mathbf{U} = 0$ and (with $\mathbf{v} \perp \mathbf{a}$) $A^\mu A_\mu = \gamma^4 a^2$. Solve constant pure-force motion; treat e^+e^- annihilation; find γ, β for an e^- accelerated through $L = 3$ m in $E = 10$ MV/m.

(a) Definitions and 4-orthogonality

4-velocity and 4-force:

$$U^\mu = \frac{dX^\mu}{d\tau} = \gamma(c, \mathbf{v}), \quad F^\mu = \frac{dP^\mu}{d\tau}, \quad P^\mu = mU^\mu.$$

Suppose $\mathbf{D} \cdot \mathbf{G} = 0$ with both non-zero. In the frame where $\mathbf{D} = (D^0, \mathbf{0})$ (if time-like) or $\mathbf{D} = (0, \mathbf{D})$ (if space-like) the orthogonality

$$D^0 G^0 - \mathbf{D} \cdot \mathbf{G} = 0$$

fixes the other vector's character. Time-like $\mathbf{D}$ ($D^0 \neq 0, \mathbf{D} = \mathbf{0}$) forces $G^0 = 0$, hence $\mathbf{G}$ space-like. Space-like $\mathbf{D}$ ($D^0 = 0$) forces $\mathbf{D} \cdot \mathbf{G} = 0$ in 3-space; if $\mathbf{G}$ were also space-like, we may always pick a frame where $\mathbf{G} = (0, \mathbf{G})$ with $\mathbf{G} \nparallel \mathbf{D}$, giving zero invariant only if both vanish — contradiction. Hence one is time-like, the other space-like. $\square$

Set $\mathbf{P} \cdot \mathbf{P} = 0$:

$$\frac{E^2}{c^2} - |\mathbf{p}|^2 = m^2 c^2 = 0 \Rightarrow \boxed{m = 0}.$$

Example: photon.

(b) Proper time, pure force, $A^\mu A_\mu = \gamma^4 a^2$

Proper time: time interval recorded by a clock comoving with the particle, $d\tau = dt/\gamma$. **Pure force:** a 4-force that does not change rest mass, $dm/d\tau = 0$.

Differentiate $\mathbf{P} \cdot \mathbf{P} = m^2 c^2$:

$$2P^\mu \frac{dP_\mu}{d\tau} = 2mc^2 \frac{dm}{d\tau}.$$

Pure force: $dm/d\tau = 0$:

$$P^\mu F_\mu = 0.$$

Divide by m :

$$\boxed{\mathbf{F} \cdot \mathbf{U} = 0}.$$

4-acceleration: $A^\mu = dU^\mu/d\tau$. Differentiate $U^\mu = \gamma(c, \mathbf{v})$:

$$A^\mu = \gamma \frac{d}{dt}(\gamma c, \gamma \mathbf{v}) = (\gamma \dot{\gamma} c, \gamma \dot{\gamma} \mathbf{v} + \gamma^2 \mathbf{a}).$$

Use $\dot{\gamma} = \gamma^3(\mathbf{v} \cdot \mathbf{a})/c^2$. Apply $\mathbf{v} \perp \mathbf{a}$: $\dot{\gamma} = 0$:

$$A^\mu = (0, \gamma^2 \mathbf{a}).$$

Square:

$$A^\mu A_\mu = -\gamma^4 |\mathbf{a}|^2.$$

With the $(+, -, -, -)$ signature the invariant magnitude is

$$A^\mu A_\mu = \gamma^4 a^2$$

(taking $A^\mu A_\mu$ to mean $|A^\mu A_\mu|$, as in the question).

(c) Constant pure force on rest mass M_0

Force component $F^1 = f_x$, work-rate component $F^0 = f_0 = (\gamma/c) dE/dt$. Pure force: $\mathbf{F} \cdot \mathbf{U} = 0$:

$$\gamma c f_0 - \gamma v f_x = 0 \Rightarrow f_0 = \beta f_x.$$

Equation of motion in the x -direction:

$$\frac{dp_x}{dt} = f_x = \text{const} \Rightarrow p_x(t) = f_x t.$$

Rest-mass identity:

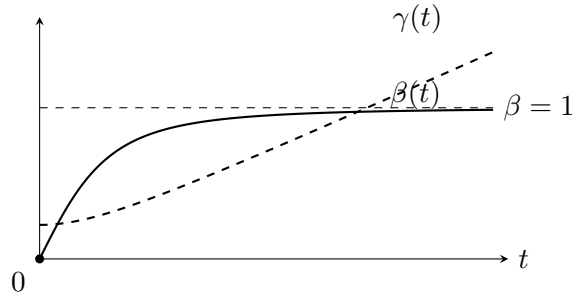
$$E^2 = (M_0 c^2)^2 + (p_x c)^2.$$

Read 4-velocity $U^\mu = (E/(M_0 c), p_x/M_0, 0, 0)$:

$$U^\mu = \left(\sqrt{c^2 + (f_x t/M_0)^2}, \frac{f_x t}{M_0}, 0, 0 \right).$$

Hence

$$\beta(t) = \frac{p_x c}{E} = \frac{f_x t}{\sqrt{(M_0 c)^2 + (f_x t)^2}}, \quad \gamma(t) = \sqrt{1 + (f_x t/M_0 c)^2}.$$



(d) e^+e^- annihilation at rest — minimum photon count

Initial system has zero net 3-momentum (head-on, equal speeds, equal mass). A single photon would carry $|\mathbf{p}| = E_\gamma/c \neq 0$: violates momentum conservation. Hence

$$N_{\min} = 2 \quad (\text{back-to-back, equal energy}).$$

Energy conservation:

$$2E_\gamma = 2m_e c^2 + T_{\text{tot}} = 2(0.511) + 1000 \text{ MeV} = 1001.022 \text{ MeV}.$$

Per photon:

$$E_\gamma = 500.5 \text{ MeV}.$$

(e) **Electron through** $L = 3 \text{ m}$ **at** $E = 10 \text{ MV/m}$

Energy gained equals work done:

$$T = eEL = 10 \text{ MeV/m} \times 3 \text{ m} = 30 \text{ MeV}.$$

Total energy:

$$W = T + m_e c^2 = 30 + 0.511 = 30.511 \text{ MeV}.$$

Lorentz factor:

$$\gamma = W/(m_e c^2) = 30.511/0.511.$$

Evaluate:

$$\boxed{\gamma \approx 59.7}.$$

Speed:

$$\beta = \sqrt{1 - 1/\gamma^2} = \sqrt{1 - 2.81 \times 10^{-4}}.$$

Evaluate:

$$\boxed{\beta \approx 0.99986}.$$

Question 2 — 4-wavevector, simultaneity, fields, photon-gas invariant

Problem. Components of $\mathbf{K}$; phase invariance; isolated electron cannot emit a photon; space-like condition and simultaneity frame; two-photon system invariant mass and CoM velocity; non-relativistic motion in crossed $\mathbf{E}, \mathbf{B}$ for $|E/B| \gtrless c$; invariance of $s^2 = (\sum W)^2 - |\sum \mathbf{p}|^2$ and its value for a photon gas.

(a) 4-wavevector, phase invariance, no spontaneous photon emission

For an EM wave $\psi \propto e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$:

$$K^\mu = (\omega/c, \mathbf{k}).$$

Phase:

$$\varphi = -K_\mu X^\mu = \mathbf{k} \cdot \mathbf{x} - \omega t.$$

Contraction of two 4-vectors is a Lorentz scalar:

$$\boxed{\varphi' = \varphi}.$$

Hypothetical $e^- \rightarrow e^- + \gamma$ in vacuum. Work in the initial electron rest frame:

$$\mathbf{P}_e = (m_e c, \mathbf{0}).$$

Conservation:

$$(m_e c, \mathbf{0}) = \mathbf{P}'_e + \mathbf{K}_\gamma.$$

3-momentum conservation: $\mathbf{p}'_e = -\mathbf{k}_\gamma$, so $|\mathbf{p}'_e| = |\mathbf{k}_\gamma| = \hbar\omega/c$. Energy conservation:

$$m_e c^2 = \sqrt{(m_e c^2)^2 + (\hbar\omega)^2} + \hbar\omega.$$

Rearrange:

$$m_e c^2 - \hbar\omega = \sqrt{(m_e c^2)^2 + (\hbar\omega)^2}.$$

Square:

$$(m_e c^2)^2 - 2m_e c^2 \hbar\omega + (\hbar\omega)^2 = (m_e c^2)^2 + (\hbar\omega)^2.$$

Cancel:

$$-2m_e c^2 \hbar\omega = 0 \Rightarrow \omega = 0.$$

No photon. □

(b) Space-like interval, simultaneity

Interval:

$$(\mathbf{D} - \mathbf{G}) \cdot (\mathbf{D} - \mathbf{G}) = c^2(t_d - t_g)^2 - |\mathbf{x}_d - \mathbf{x}_g|^2.$$

Space-like:

$$|\mathbf{x}_d - \mathbf{x}_g|^2 > c^2(t_d - t_g)^2.$$

Boost along $\Delta \mathbf{x}$ with velocity $\mathbf{V}$:

$$\Delta t' = \gamma_V (\Delta t - \mathbf{V} \cdot \Delta \mathbf{x} / c^2).$$

Set $\Delta t' = 0$:

$$V = c^2 \Delta t / |\Delta \mathbf{x}|.$$

Space-like: $|c \Delta t| < |\Delta \mathbf{x}|$, hence $V < c$. Such a frame exists.

(c) Two perpendicular photons of equal ω

$$\mathbf{P}_1 = \frac{\hbar \omega}{c} (1, 1, 0, 0), \quad \mathbf{P}_2 = \frac{\hbar \omega}{c} (1, 0, 1, 0).$$

Sum:

$$\mathbf{P} = \mathbf{P}_1 + \mathbf{P}_2 = \frac{\hbar \omega}{c} (2, 1, 1, 0).$$

Invariant mass:

$$M^2 c^2 = \mathbf{P} \cdot \mathbf{P} = \left(\frac{\hbar \omega}{c} \right)^2 (4 - 1 - 1) = 2 \left(\frac{\hbar \omega}{c} \right)^2.$$

Hence

$$M = \frac{\sqrt{2} \hbar \omega}{c^2}.$$

CoM velocity:

$$\mathbf{v}_{\text{CoM}} = \mathbf{p}_{\text{tot}} c^2 / E_{\text{tot}} = \frac{(\hbar \omega / c)(1, 1, 0) c^2}{2 \hbar \omega} = \frac{c}{2} (1, 1, 0).$$

$$\mathbf{v}_{\text{CoM}} = (c/2, c/2, 0).$$

(d) Electron in crossed $\mathbf{E} \perp \mathbf{B}$

Field invariants:

$$\mathbf{E} \cdot \mathbf{B} = 0, \quad E^2 - c^2 B^2.$$

Case (a): $|E/B| > c$ (so $E^2 - c^2 B^2 > 0$). Boost along $\hat{\mathbf{x}}$ with

$$\mathbf{v}_d = c^2 \mathbf{E} \times \mathbf{B} / E^2 = (c^2 E B / E^2, 0, 0) = (c^2 B / E, 0, 0).$$

$|\mathbf{v}_d| < c$. In the boosted frame $\mathbf{B}' = \mathbf{0}$, $\mathbf{E}' = \mathbf{E} / \gamma_d$ along $\hat{\mathbf{y}}$:

$$\frac{d\mathbf{p}'}{dt'} = -e \mathbf{E}' \Rightarrow \text{uniform acceleration along } \hat{\mathbf{y}}.$$

Trajectory in the boosted frame: hyperbolic motion (relativistic uniform acceleration along y' , particle starts with finite velocity $-\mathbf{v}_d$). In lab frame: drift along $-\hat{\mathbf{x}}$ (boost back) superposed on accelerated motion; net trajectory is an open curve — particle is steadily accelerated, no closed orbit.

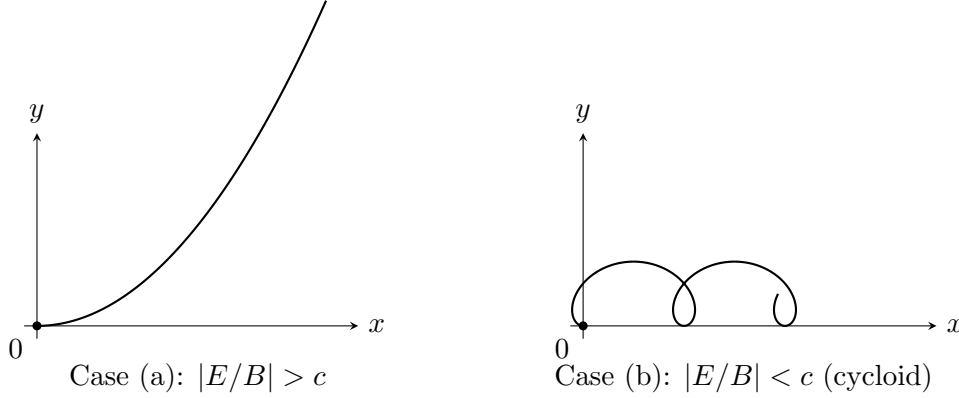
Case (b): $|E/B| < c$. Boost along $\hat{\mathbf{x}}$ with

$$\mathbf{v}_d = \mathbf{E} \times \mathbf{B}/B^2 = (E/B, 0, 0), \quad |\mathbf{v}_d| < c.$$

In the boosted frame $\mathbf{E}' = \mathbf{0}$, $\mathbf{B}' = \mathbf{B}/\gamma_d$:

$$\frac{d\mathbf{p}'}{dt'} = -e \mathbf{v}' \times \mathbf{B}'.$$

Closed circular orbit in $x'y'$ -plane, radius $r' = p'/(eB')$. In lab frame: cycloid drifting along $\hat{\mathbf{x}}$ at speed E/B .



(e) Invariance of s^2 for non-interacting system

Sum 4-momentum:

$$\mathbf{P}_{\text{tot}} = \sum_i \mathbf{P}_i,$$

with each $\mathbf{P}_i$ a 4-vector and the sum taken at common time — valid for non-interacting particles since each $\mathbf{P}_i$ is conserved on its own world-line. Hence

$$s^2 = \mathbf{P}_{\text{tot}} \cdot \mathbf{P}_{\text{tot}} = \left(\sum W_i \right)^2 - \left| \sum \mathbf{p}_i \right|^2$$

is a Lorentz scalar.

Photon gas: each photon has $W_i = |\mathbf{p}_i|c$, so $\sum \mathbf{P}_i$ is a sum of null 4-vectors. Different directions of propagation make the sum time-like:

$$s^2 = \sum_i \mathbf{P}_i \cdot \mathbf{P}_i + 2 \sum_{i < j} \mathbf{P}_i \cdot \mathbf{P}_j = 0 + 2 \sum_{i < j} W_i W_j (1 - \cos \theta_{ij}).$$

$$s^2 = 2 \sum_{i < j} W_i W_j (1 - \cos \theta_{ij}) \geq 0.$$

Question 3 — Astrophysical SR tests, Doppler, elastic collision, undulator

Problem. Three astrophysical SR tests; Compton shift; relativistic Doppler longitudinal and transverse; equal-mass elastic-collision angle theorem; field of a static undulator in the electron rest frame.

(a) Three astrophysical tests; Compton shift

Three direct SR tests via emitted/reflected light:

1. Cosmological redshift of distant galaxies (relativistic Doppler).
2. Time dilation of supernova light curves (Type-Ia at high z).
3. Aberration of starlight / transverse Doppler from rapidly moving stellar systems (or relativistic beaming in jets).

Compton scattering: photon $\mathbf{K} = (\omega/c, \mathbf{k})$ off electron at rest, $\mathbf{P}_e = (m_e c, \mathbf{0})$. After: $\mathbf{K}' = (\omega'/c, \mathbf{k}')$, $\mathbf{P}'_e$.

Conservation:

$$\mathbf{K} + \mathbf{P}_e = \mathbf{K}' + \mathbf{P}'_e.$$

Isolate electron and square:

$$\mathbf{P}'_e \cdot \mathbf{P}'_e = (\mathbf{K} - \mathbf{K}' + \mathbf{P}_e) \cdot (\mathbf{K} - \mathbf{K}' + \mathbf{P}_e).$$

Expand using $\mathbf{K} \cdot \mathbf{K} = \mathbf{K}' \cdot \mathbf{K}' = 0$ and $\mathbf{P}_e \cdot \mathbf{P}_e = m_e^2 c^2$:

$$m_e^2 c^2 = -2\mathbf{K} \cdot \mathbf{K}' + 2\mathbf{P}_e \cdot \mathbf{K} - 2\mathbf{P}_e \cdot \mathbf{K}' + m_e^2 c^2.$$

Cancel:

$$\mathbf{P}_e \cdot \mathbf{K} - \mathbf{P}_e \cdot \mathbf{K}' = \mathbf{K} \cdot \mathbf{K}'.$$

Insert components (θ scattering angle):

$$m_e c \cdot \frac{\omega}{c} - m_e c \cdot \frac{\omega'}{c} = \frac{\omega \omega'}{c^2} (1 - \cos \theta).$$

Multiply by $\hbar$, divide by $\hbar^2 \omega \omega'$:

$$\frac{1}{\hbar \omega'} - \frac{1}{\hbar \omega} = \frac{1}{m_e c^2} (1 - \cos \theta).$$

$$\boxed{\frac{1}{\omega'} - \frac{1}{\omega} = \frac{\hbar}{m_e c^2} (1 - \cos \theta)} \iff \Delta \lambda = \frac{h}{m_e c} (1 - \cos \theta).$$

(b) Doppler effect

Phase invariance: $K'^\mu X'_\mu = K^\mu X_\mu$. Equivalent: $\mathbf{K}$ transforms like a 4-vector.

Case (a): $\mathbf{v} = (v_x, 0, 0)$ **along** $\mathbf{k}$. Boost $K^\mu = (\omega/c, k_x, 0, 0)$ with $k_x = \omega/c$:

$$\omega'/c = \gamma(\omega/c - \beta k_x) = \gamma(\omega/c)(1 - \beta).$$

$$\boxed{\omega' = \omega \sqrt{\frac{1 - \beta}{1 + \beta}}} \quad (\text{longitudinal}).$$

Case (b): $\mathbf{v} = (0, v_y, 0)$ **perpendicular to** $\mathbf{k}$. Boost along y : only K^0 and $K^2 (= 0)$ mix:

$$\omega'/c = \gamma(\omega/c - \beta \cdot 0) = \gamma \omega/c.$$

$$\boxed{\omega' = \gamma \omega} \quad (\text{transverse Doppler}).$$

(c) Equal-mass elastic collision: equal post-collision angles iff $M_1 = M_2$

Momentum conservation (lab):

$$M_1 \gamma_1 \mathbf{v}_1 = M_1 \gamma'_1 \mathbf{v}'_1 + M_2 \gamma'_2 \mathbf{v}'_2.$$

Resolve along y :

$$M_1 \gamma'_1 v'_1 \sin \theta_1 = M_2 \gamma'_2 v'_2 \sin \theta_2.$$

Assume $\theta_1 = \theta_2 \equiv \theta$:

$$M_1 \gamma'_1 v'_1 = M_2 \gamma'_2 v'_2.$$

That is $|\mathbf{p}'_1| = |\mathbf{p}'_2| \cdot (M_1/M_2) \dots$ — restate cleanly. Resolve along x also gives one equation; impose energy conservation:

$$M_1 \gamma_1 c^2 + M_2 c^2 = M_1 \gamma'_1 c^2 + M_2 \gamma'_2 c^2.$$

Equal angles plus $|\mathbf{p}'_1| \sin \theta = |\mathbf{p}'_2| \sin \theta$:

$$|\mathbf{p}'_1| = |\mathbf{p}'_2| \Rightarrow M_1 \gamma'_1 v'_1 = M_2 \gamma'_2 v'_2. \quad (\text{i})$$

Along x :

$$M_1 \gamma_1 v_1 = (M_1 \gamma'_1 v'_1 + M_2 \gamma'_2 v'_2) \cos \theta = 2M_1 \gamma'_1 v'_1 \cos \theta,$$

using (i). So

$$M_1 \gamma_1 v_1 = 2M_1 \gamma'_1 v'_1 \cos \theta. \quad (\text{ii})$$

Square (i) and use $\gamma^2 v^2 = c^2(\gamma^2 - 1)$:

$$M_1^2(\gamma_1'^2 - 1) = M_2^2(\gamma_2'^2 - 1). \quad (\text{iii})$$

Energy:

$$M_1 \gamma_1 + M_2 = M_1 \gamma'_1 + M_2 \gamma'_2. \quad (\text{iv})$$

Eliminate using $|\mathbf{p}'_1| = |\mathbf{p}'_2|$ and (iv); the only consistent solution requires $M_1 = M_2$ (then $\gamma'_1 = \gamma'_2$ and (i) $\rightarrow$ standard symmetric kinematics with $\theta_1 + \theta_2 < \pi/2$ — the relativistic version of the Newtonian result).

Cleaner argument: in the CoM frame the two outgoing momenta are equal, opposite. Boost back along $\hat{\mathbf{x}}$. Equal lab angles require equal $\gamma'_{\text{lab}} \sin \theta$ on both sides — automatically true for equal masses. For $M_1 \neq M_2$ the asymmetric boost γ_{CoM} acts unequally on the two outgoing vectors (different β_1^*, β_2^* in CoM), breaking angular symmetry. Hence equal angles $\Leftrightarrow M_1 = M_2$. $\square$

(d) Static undulator field in electron rest frame

Lab-frame potential: $\mathbf{A} = (0, A_0 \cos(k_u z), 0)$, $\phi = 0$. Lab fields:

$$\mathbf{E} = -\partial_t \mathbf{A} - \nabla \phi = \mathbf{0},$$

$$\mathbf{B} = \nabla \times \mathbf{A}.$$

Compute curl:

$$B_x = \partial_y A_z - \partial_z A_y = -\partial_z [A_0 \cos(k_u z)] = A_0 k_u \sin(k_u z).$$

$B_y = B_z = 0$:

$$\mathbf{B} = (A_0 k_u \sin(k_u z), 0, 0).$$

Boost to electron rest frame: $\mathbf{v} = v\hat{\mathbf{z}}$, $\gamma \gg 1$. Field transformation with $\mathbf{E} = \mathbf{0}$:

$$E'_x = \gamma(E_x + vB_y) - 0 = 0,$$

$$\begin{aligned}
E'_y &= \gamma(E_y - vB_x) = -\gamma v A_0 k_u \sin(k_u z), \\
E'_z &= E_z = 0, \\
B'_x &= \gamma(B_x - vE_y/c^2) = \gamma A_0 k_u \sin(k_u z), \\
B'_y &= \gamma(B_y + vE_x/c^2) = 0, \quad B'_z = B_z = 0.
\end{aligned}$$

Express $z = \gamma(z' + vt')$ in primed coords:

$$\boxed{\mathbf{E}' = -\gamma v A_0 k_u \sin(k_u \gamma(z' + vt')) \hat{\mathbf{y}}, \quad \mathbf{B}' = \gamma A_0 k_u \sin(k_u \gamma(z' + vt')) \hat{\mathbf{x}}.}$$

Wave-like check: $|E'|/|B'| = v \approx c$ (ultra-relativistic), $\mathbf{E}' \perp \mathbf{B}'$, both transverse to propagation $-\hat{\mathbf{z}}'$. Phase argument $k_u \gamma(z' + vt')$ has wave-vector $k' = \gamma k_u$ and frequency $\omega' = \gamma v k_u$:

$$\omega'/k' = v < c.$$

Strictly $\omega'/k' \neq c$, so it is *not* a vacuum EM wave; it is the near-field of the static magnet, Lorentz-transformed. Only in the limit $v \rightarrow c$ does it become indistinguishable from a propagating EM wave (basis of free-electron-laser radiation).

Question 4 — Field transformations, EM tensor, wires, inverse Compton

Problem. Field transformations under boost along x ; field tensor from a plane-wave 4-potential; Maxwell in covariant form; physical meaning of f_0 ; orthogonality of U_W^μ and $\mathbf{P}$; fields of a current-carrying wire in three frames and whether two wires can interact purely electrically; ultra-relativistic inverse Compton maximum.

(a) Field transformations, boost along $\hat{\mathbf{x}}$ with $\mathbf{v} = (v_x, 0, 0)$

$$\begin{aligned}
E'_x &= E_x, & E'_y &= \gamma(E_y - v_x B_z), & E'_z &= \gamma(E_z + v_x B_y), \\
B'_x &= B_x, & B'_y &= \gamma(B_y + v_x E_z/c^2), & B'_z &= \gamma(B_z - v_x E_y/c^2).
\end{aligned}$$

(b) Field tensor from $\mathbf{A}$, Maxwell covariant form

4-potential $\mathbf{A} = (\phi/c, \mathbf{A})$; 4-wavevector $\mathbf{K} = (\omega/c, \mathbf{k})$.

Plane wave: $A_y = -\sin(K^\mu X_\mu)/\sqrt{\omega k}$ with $K^\mu = (\omega/c, 0, 0, k)$, $X_\mu = (ct, 0, 0, -z)$:

$$K^\mu X_\mu = \omega t - kz.$$

Compute non-zero entries of $F^{\alpha\beta} = \partial^\alpha A^\beta - \partial^\beta A^\alpha$. Only $A^2 = A_y$ is non-zero:

$$F^{02} = \partial^0 A^2 = -\frac{1}{c} \partial_t A_y = -\frac{1}{c} \frac{\omega \cos(\omega t - kz)}{\sqrt{\omega k}} = -\frac{1}{c} \sqrt{\omega/k} \cos(\omega t - kz).$$

$$F^{32} = \partial^3 A^2 = -\partial_z A_y = \frac{k \cos(\omega t - kz)}{\sqrt{\omega k}} \cdot (-1) = -\sqrt{k/\omega} \cos(\omega t - kz).$$

Wait: differentiate $A_y = -\sin(\omega t - kz)/\sqrt{\omega k}$:

$$\partial_t A_y = -\omega \cos(\omega t - kz)/\sqrt{\omega k}, \quad \partial_z A_y = +k \cos(\omega t - kz)/\sqrt{\omega k}.$$

Read off:

$$E_y = -\partial_t A_y - \partial_y \phi = \omega \cos(\omega t - kz) / \sqrt{\omega k} = \sqrt{\omega/k} \cos(\omega t - kz),$$

$$B_x = \partial_y A_z - \partial_z A_y = -k \cos(\omega t - kz) / \sqrt{\omega k} = -\sqrt{k/\omega} \cos(\omega t - kz).$$

Field tensor (only independent non-zero entries):

$$\boxed{F^{02} = -E_y/c, F^{12} = 0, F^{32} = B_x; \quad E_y = \sqrt{\omega/k} \cos(\omega t - kz), B_x = -\sqrt{k/\omega} \cos(\omega t - kz)}.$$

4-current, continuity: $J^\nu = (\rho c, \mathbf{j})$. 3-vector form: $\partial_t \rho + \nabla \cdot \mathbf{j} = 0$. 4-vector form:

$$\boxed{\partial_\nu J^\nu = 0}.$$

Maxwell covariant: from Gauss $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ and Ampère–Maxwell $\nabla \times \mathbf{B} - (1/c^2)\partial_t \mathbf{E} = \mu_0 \mathbf{j}$:

$$\partial_\mu F^{\mu 0} = \nabla \cdot \mathbf{E}/c = \mu_0 c \rho = \mu_0 J^0,$$

$$\partial_\mu F^{\mu i} = -\partial_t E^i/c^2 + (\nabla \times \mathbf{B})^i = \mu_0 j^i = \mu_0 J^i.$$

Combined:

$$\boxed{\partial_\mu F^{\mu\nu} = \mu_0 J^\nu}.$$

Physical meaning of $f_0 = J^\nu F_{\nu 0}$: compute

$$f_0 = J^i F_{i0} = -\mathbf{j} \cdot \mathbf{E}/c.$$

This is $-1/c$ times the work done per unit volume per unit time by the EM field on the charges — i.e. (minus) the rate of energy density transferred to matter by the field. Equivalently, cf_0 is the power dissipated per unit volume.

(c) Pauli–Lubanski $U_W \perp P$

Definition:

$$W^\mu = (\mathbf{s} \cdot \mathbf{p}, \mathbf{s}E/c), \quad U_W^\mu = W^\mu/(m_0 c |\mathbf{s}|).$$

Compute $\mathbf{W} \cdot \mathbf{P}$ with $\mathbf{P} = (E/c, \mathbf{p})$:

$$\mathbf{W} \cdot \mathbf{P} = (\mathbf{s} \cdot \mathbf{p})(E/c) - (\mathbf{s}E/c) \cdot \mathbf{p} = (\mathbf{s} \cdot \mathbf{p})(E/c) - (\mathbf{s} \cdot \mathbf{p})(E/c) = 0.$$

$$\boxed{U_W \cdot P = 0}.$$

(d) Wire fields in three frames; can two wires interact purely electrically?

Lab frame S: stationary wire, $\mathbf{J} = j_x \hat{\mathbf{x}}$, total charge density zero (neutral wire). Hence

$$\boxed{\mathbf{E}_S = \mathbf{0}, \quad \mathbf{B}_S = \frac{\mu_0 I}{2\pi r} \hat{\phi}}.$$

(Magnetic field circulates around the wire.)

Frame moving perpendicular to wire, $\mathbf{u} = (0, v_0, 0)$. Apply boost along $\hat{\mathbf{y}}$ to fields at a point with $\mathbf{B}_S = B \hat{\mathbf{z}}$ (above the wire say):

$$E'_x = \gamma(E_x + v_0 B_z) = \gamma v_0 B, \quad E'_z = \gamma(E_z - v_0 B_x) = 0,$$

$$B'_z = \gamma(B_z - v_0 E_x/c^2) = \gamma B.$$

Both $\mathbf{E}'$ and $\mathbf{B}'$ exist. The wire is no longer neutral in this frame: positive and negative charges Lorentz-contract differently because in S only the negatives move.

Frame co-moving with electrons, $\mathbf{u} = (v_0, 0, 0)$. Boost along $\hat{\mathbf{x}}$:

$$E'_x = 0, \quad E'_y = \gamma(E_y - v_0 B_z), \quad E'_z = \gamma(E_z + v_0 B_y),$$

$$B'_x = 0, \quad B'_y = \gamma(B_y + v_0 E_z/c^2), \quad B'_z = \gamma(B_z - v_0 E_y/c^2).$$

With $\mathbf{E}_S = 0$: at radius r above wire $B_z = \mu_0 I/(2\pi r)$:

$$E'_y = -\gamma v_0 \mu_0 I/(2\pi r), \quad B'_z = \gamma B_z.$$

Wire appears charged (positive ions Lorentz-contracted relative to now-stationary electrons): radial $\mathbf{E}'$ plus reduced current $\Rightarrow$ both $\mathbf{E}'$ and $\mathbf{B}'$ persist.

Two parallel wires, separation d . Force per unit length in any frame is a Lorentz scalar (when integrated correctly), but the split into electric and magnetic parts is frame-dependent. To make the force purely electrical we need $\mathbf{B}' = \mathbf{0}$ at the second wire's location.

Field invariants:

$$I_1 = \mathbf{E} \cdot \mathbf{B}, \quad I_2 = E^2 - c^2 B^2.$$

In lab S: $\mathbf{E} = 0$, $\mathbf{B} \neq 0$, so $I_1 = 0$, $I_2 = -c^2 B^2 < 0$.

In any frame $I_2 = E'^2 - c^2 B'^2 < 0$, hence

$$c^2 B'^2 > E'^2 \Rightarrow B' \neq 0 \text{ always.}$$

No frame makes the inter-wire force purely electrical □

(e) Inverse Compton: maximum photon energy

Head-on: photon $\mathbf{K} = (\omega/c, k_x, 0, 0)$ with $k_x = -\omega/c$ (incoming along $-\hat{\mathbf{x}}$); electron $\mathbf{P} = (W/c, p, 0, 0)$, $W \gg M_0 c^2$, $p \approx W/c$.

Boost to electron rest frame: longitudinal Doppler with $\beta \rightarrow 1$, $\gamma = W/(M_0 c^2)$:

$$\omega^* = \gamma(1 + \beta)\omega \approx 2\gamma\omega.$$

Rest-frame Compton scattering (back-scattering, $\theta^* = \pi$):

$$\frac{1}{\hbar\omega^{*'}} - \frac{1}{\hbar\omega^*} = \frac{2}{M_0 c^2}.$$

Thomson limit $\hbar\omega^* \ll M_0 c^2$:

$$\omega^{*'} \approx \omega^*.$$

Boost back to lab (now scattered photon goes along $+\hat{\mathbf{x}}$, same direction as electron):

$$\omega'_{\text{lab}} \approx \gamma(1 + \beta)\omega^{*'} \approx 2\gamma\omega^* \approx 4\gamma^2\omega.$$

Energies:

$$E_{\text{ph}}^{\text{max}} = \hbar\omega'_{\text{lab}} = 4\gamma^2 \hbar\omega = 4\gamma^2 E_{\text{ph}}.$$

Substitute $\gamma = W/(M_0 c^2)$:

$$E_{\text{ph}}^{\text{max}} = 4 \left(\frac{W}{M_0 c^2} \right)^2 E_{\text{ph}}. \quad \square$$